



Integrated College ERP Dashboard & Distributed Academic Management System

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Abstract

- A unified, role-based educational ERP uniting a 21-table normalized MySQL database with distributed Node.js REST APIs and a React 19 interface.
- Outcome: Automates attendance, grade entry, fee clearance and hall tickets with sub-100ms response.

Problem Statement & Objectives

Problem: Decentralized spreadsheets and paper slips cause high data redundancy, scheduling conflicts, and slow manual clearances.

Objective 1: Design a normalized 3NF MySQL relational database with 21 tables and cascade integrity.

Objective 2: Build secure REST APIs with JWT role-based access control (RBAC) and ACID compliance.

Objective 3: Implement real-time workflows for attendance tracking, fee invoicing, and digital No Dues verification.

Scope: 3 roles (Admin, Faculty, Student) across 15+ modules (Academics, Exams, Hostel, Library, Transport).

Existing vs Proposed System

Feature	Existing System	Proposed System
Data Store	Isolated spreadsheets and data silos	Centralized 3NF MySQL relational database
Authentication	Insecure plain text or basic sessions	Encrypted JWT token-based Role-Based Access Control
Attendance Tracking	Paper registers with delayed manual entry	Live matrix logger with automatic <75% shortage alert badges
No Dues Clearance	Physical paper signature runs across departments	Instant 1-click atomic digital clearance validation
User Onboarding	Slow manual single-user registration	High-throughput batch Excel (.xlsx) ingestion with auto-rollback

Feasibility Study

- Technical: Standard open-source stack (React 19, Node.js, MySQL 8.0) with extensive documentation.
- Economic: Zero software licensing cost using open-source tools and free-tier cloud deployment.
- Operational: Intuitive, responsive role-based UI requiring no technical training for students or faculty.
- Schedule: Database, backend APIs and frontend integration completed within the PBL semester.

Results & Testing

- 100% CRUD pass across 21 tables; zero orphan records on cascade delete.
- 100% Postman test pass; sub-100ms average query time.

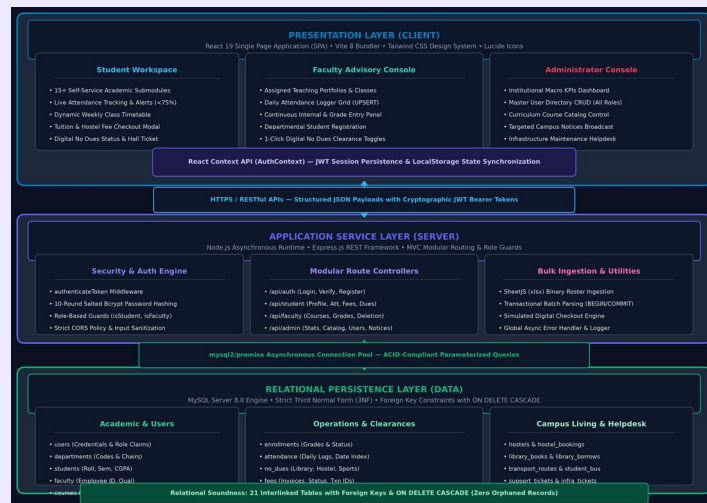
Conclusion

- Replaced fragmented paper-based university processes with an automated digital platform.
- Met all 25CS1302E objectives with scalable REST APIs on an ACID-compliant database.

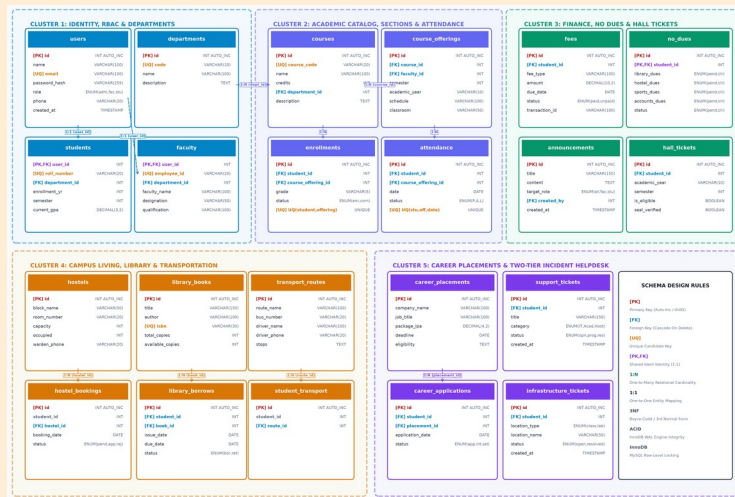
Tools & Technologies

- Front-End: React 19, Vite, Tailwind CSS, Lucide-React
- Back-End: Node.js, Express.js, REST APIs, JWT, BcryptJS
- Database: MySQL 8.0 (InnoDB engine, 21 relational tables, ACID compliant)
- Tools: Git & GitHub, Postman, VS Code, Draw.io

System Architecture



ER Diagram & Database Design



Implementation Screenshots

